Introduction to Database – TD01

Topic

Database models and relational algebra: hierarchical, network, relational model design, and querying with algebra operators.

Group Setup

- Group size: **2 or 3 students**
- Time: **90 minutes**
- Submit: **1 PDF per group**
- File name: `TD01_GroupXX.pdf`

Resources to Use

- `Lecture/Lesson 2 - Relational Algebra.pdf`
- `Lecture/Lesson 3 - Database Model.pdf`

Learning Goals

- Apply the hierarchical model to represent tree-structured data.
- Apply the network model to represent many-to-many relationships.
- Design a relational schema from textual requirements.
- Write relational algebra expressions to query a given schema.

Part A – Hierarchical Model

A school wants to have a database to manage information of student, teacher and course. Student is identified by ID. The database needs to store name, date of birth and gender of student. Teacher is identified by a unique number. Other information of teacher is name, skill, gender and degree. Course is identified by unique code. Course has title and description. Student takes courses. A course can have many students. A teacher can teach one or many courses. A course is given by only one teacher. Each student has a score for each course.

Design using **hierarchical model** for the above system.

Part B – Network Model

A company database needs to store information about employees (identified by ssn, with salary and phone as attributes), departments (identified by dnu, with dname and budget as attributes), and children of employees (with name and age as attributes). Employees work in departments; each department is managed by an employee; a child must be identified uniquely by name when the parent (who is an employee; assume that only one parent works for the company) is known. We are not interested in information about a child once the parent leaves the company.

Design using **network model** for the database above.

Part C – Relational Model

The description of a system is represented as following:

- An author is identified by a reference (ID). We can search for some information related to each author such as his name and his address.
- A book is identified by an ISBN. We store also the title, and the price of each book.
- An editor is identified by an editor's number. For each editor, it is defined by his name and his address.
- A store is identified by a store's number. For each store, we can know its name, its address, its contact such as the telephones' numbers, email address or web site. We know also the number of unit for each book in the stock.
- A collection is identified by a name of collection and a short description.
- An author in our system writes at least one book and can write also many of them.
- A book is written by one or more authors.
- A book is published by only one editor.
- Each editor publishes many books.
- All the books are sold in several stores.
- All the stores sell many different books.
- An editor can classify the books as collections.
- A book belongs to only one collection.
- A collection is named by only one editor.
- A collection always contains many books.

Design using **relational model** for the above system.

Part D – Relational Algebra

Use the relational schema your group designed in Part C.

Write a relational algebra expression for each query:

1. Show the titles of all books.
2. Show the names of all authors whose address is `'Phnom Penh'`.
3. Show the titles of books that cost more than $20.
4. Show the names of authors who wrote the book with ISBN `'978-0-13'`.
5. Show all pairs of (author name, book title) for actual authorship.
6. Show the titles of books published by editor `E01`.
7. Show the names of stores that sell the book `'978-0-13'`.
8. Show the titles of books that are **not** sold in any store.

Evaluation (10 points)

- Part A – Hierarchical model: 2 points
- Part B – Network model: 2 points
- Part C – Relational model: 3 points
- Part D – Relational algebra: 3 points